

INSY6212 - Assignment 2

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Group Members

Name	Student Number
Dylan Olsen	ST10440133
Tyreece Sewbaran	ST10439074
Ndumiso Jwara	ST10440124
Ethan Edwards	ST10442420
David Forster	ST10247775

Question 1

Question 1.1

Importance of a Project Evaluation Form

An organized method for evaluating and rating possible projects prior to their official selection and start is a project evaluation form. It is crucial because it gives decision-makers an unbiased, data-driven foundation, guaranteeing that the project selected will maximize return on investment and be in line with strategic objectives **(APM, 2023)**.

Project evaluation forms prevent projects from being chosen based on bias or intuition. It compels a methodical assessment of how effectively the SOS app concept satisfies community needs in comparison to alternative concepts. By clearly contrasting the anticipated benefits—such as increased community safety and maybe saved lives—against the expenses, the form aids in the justification of the substantial resources committed. For the team to determine whether the project is worthwhile despite these obstacles, a Project Evaluation Form is required to highlight potential risks such as network reliability in rural areas and data privacy concerns **(APM, 2023)**.

Question 1.2

What is Cross-Platform Development?

Using a single codebase to produce an application that can operate on various mobile operating systems—mostly Android and iOS—is known as cross-platform development. Developers utilize frameworks like React Native, Flutter, or Xamarin to write the code once and publish it everywhere, preserving a near-native look, feel, and performance, as opposed to creating two distinct native apps (one in Swift for iOS and one in Java/Kotlin for Android) **(TechTarget, 2024)**.

How will Cross-Platform Development Contribute to Meeting Your Project Timelines?

As a result, development time will be shortened. Compared to creating two native apps, the initial development effort is essentially cut in half because only one set of basic functionalities (SOS button, location sharing, and contact notification) need to be implemented. After deployment, only one codebase needs to be updated with any problem fixes or feature additions, and both platforms will then see the changes. By guaranteeing that iOS and Android customers receive upgrades at the same time, this expedites the post-launch support phase and saves a significant amount of time during the project's 12-month lifespan (**Warp Development, 2024**).

Three potential risks associated with cross-platform development

1. Cross-platform programs occasionally perform marginally worse than native apps when it comes to extremely sophisticated, graphics-intensive, or processor-heavy operations. Advanced capabilities like real-time augmented reality overlays may be difficult to implement, even while the basic SOS functionality is supported properly.
2. The reliability and durability of the selected cross-platform framework (such as Flutter) is critical to the project's success. The team must rely on the quality and availability of a third-party plugin, which might not be as reliable or well-maintained as native APIs, if a particular device feature is required.
3. Native developers have instant access to new features when Google or Apple release an OS update. The integration of the newest platform capabilities into the SOS app may be delayed since cross-platform developers must wait for the framework (such as React Native) to upgrade its bridge to accommodate those new features.

(Warp Development, 2024)

Explain how Cross-Platform Development will Enhance the User Experience and Usability of your Application.

The user interface and experience will be the same for all users, regardless of whether they are using iOS or Android. In an emergency app where confusion must be kept to a minimum, this consistency is essential. Switching phone brands won't need a user to relearn how to send an SOS signal, making the experience more dependable and predictable **(Edwards, 2024)**.

The app instantly becomes available to almost every smartphone user in the community by launching on both main platforms at the same time on day one. This universal accessibility is essential to the app's main goal, which is to maximize its usability and public safety impact by making sure that no one is left without help because of their device's operating system **(Edwards, 2024)**.

Question 1.3

Project Purpose & Business Case

The project's existence is formally justified by this deliverable. It will highlight the community's main safety worry, which is the absence of a quick, centralized method for calling for help in an emergency. The business case will list the concrete and intangible advantages, like quicker emergency response times, heightened resident security, and the possibility of saving lives. It acts as the cornerstone document that outlines the reasons why the R1.75 million investment is both essential and beneficial for the community's well-being, directly relating to the main objective of the technology program **(Gillis, 2025)**.

Project Scope & Key Deliverables

This part establishes the project's parameters, clearly outlining what will and won't be included to avoid "scope creep." The project will include the creation of a cross-platform mobile application called the "Guardian-SOS" app, which will have essential features such as an emergency contact notification system, GPS/SMS position sharing, and a one-tap SOS button. Importantly, to set clear expectations for stakeholders, it will also list exclusions, such as the production of any physical hardware devices, integration with official citywide emergency service dispatches, or the provision of on-the-ground medical assistance **(Gillis, 2025)**.

Milestone Schedule

The 12-month project is broken down into important, quantifiable phases in this deliverable, which offers a timeline overview. It will involve significant achievements like "Completion of UX/UI Wireframes and Prototypes by End of Month 2," "Core App Development and Alpha Testing Completed by Month 6," "Public Beta Release and User Acceptance Testing (UAT) by Month 9," along with "Final App Launch on Google Play and Apple App Store by Month 12." This timetable guarantees that progress can be monitored monthly and offers distinct benchmarks for stakeholder evaluations **(Gillis, 2025)**.

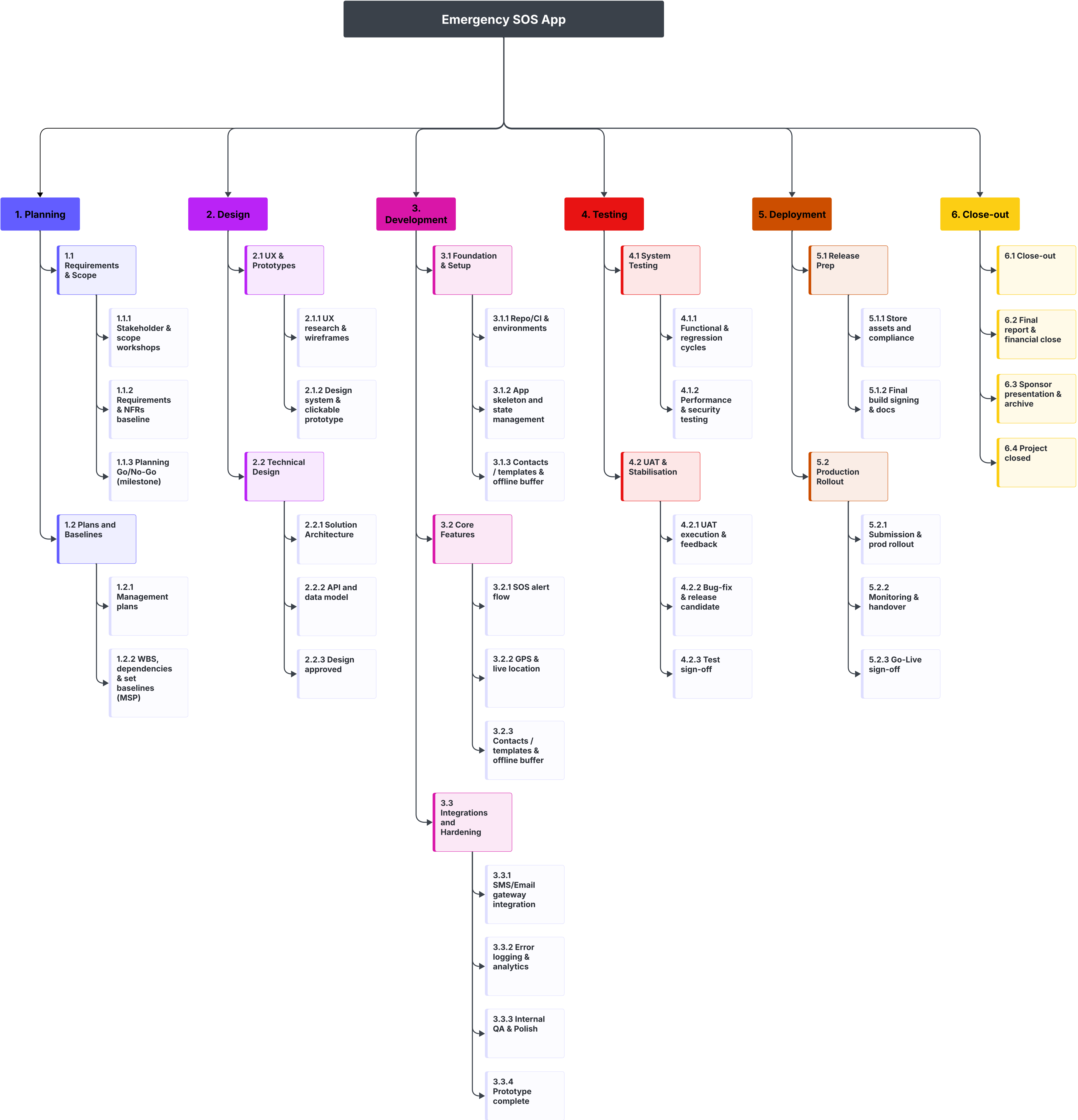
Budget & Resource Allocation

This gives the core team structure and financial plan approval. It will verify that the R1,750,000 fixed budget has been allotted to the following important areas: team salaries, software licenses (such as design tools and development frameworks), testing equipment, and a contingency fund. Additionally, it will legally approve the creation of the core project team, defining the four essential roles (e.g., Project Manager, Lead Cross-Platform Developer, UX/UI Designer, and QA Tester) and giving the project manager the power to appoint people to these roles **(Gillis, 2025)**.

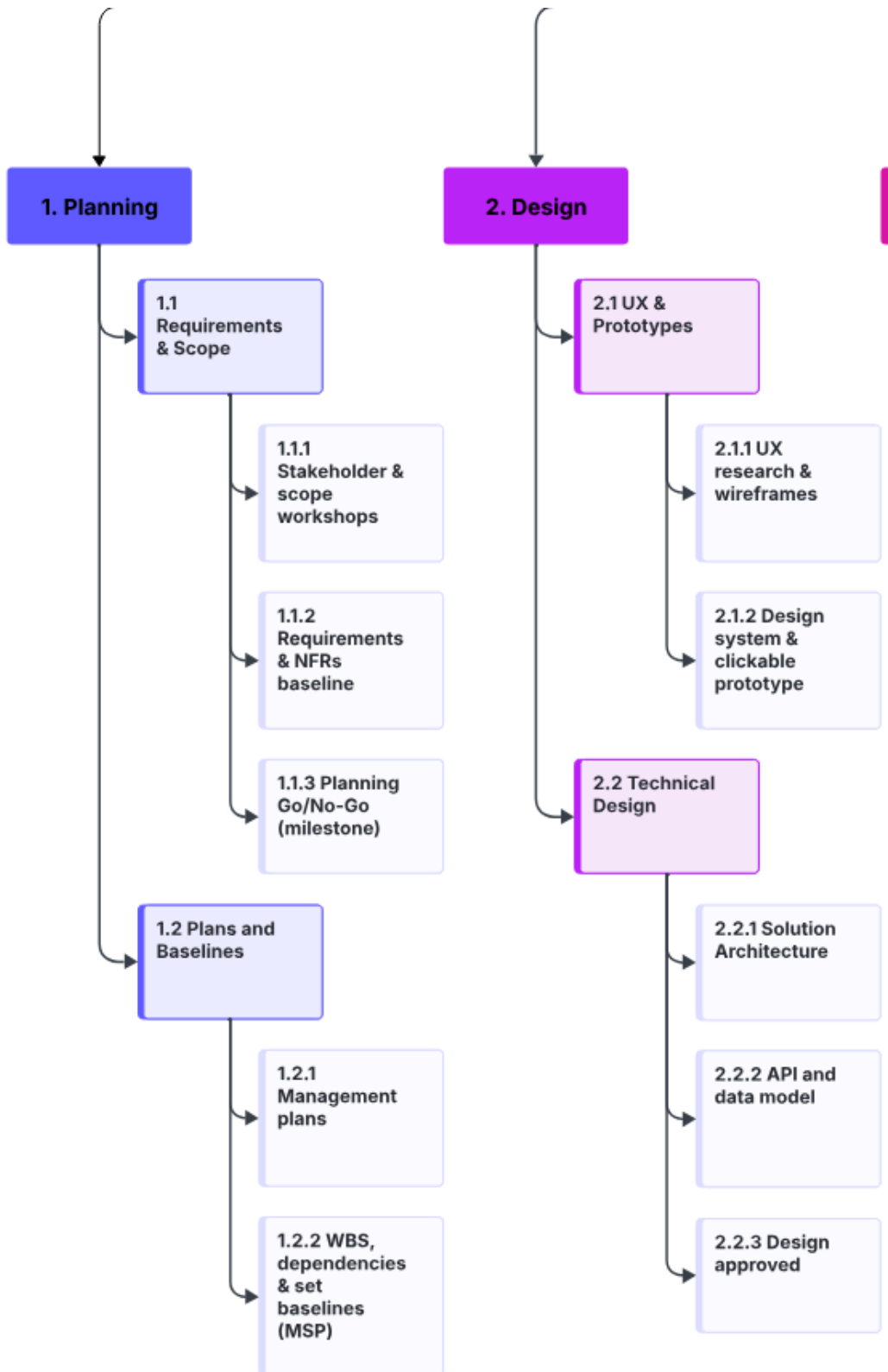
Project Manager Appointment & Authority Level

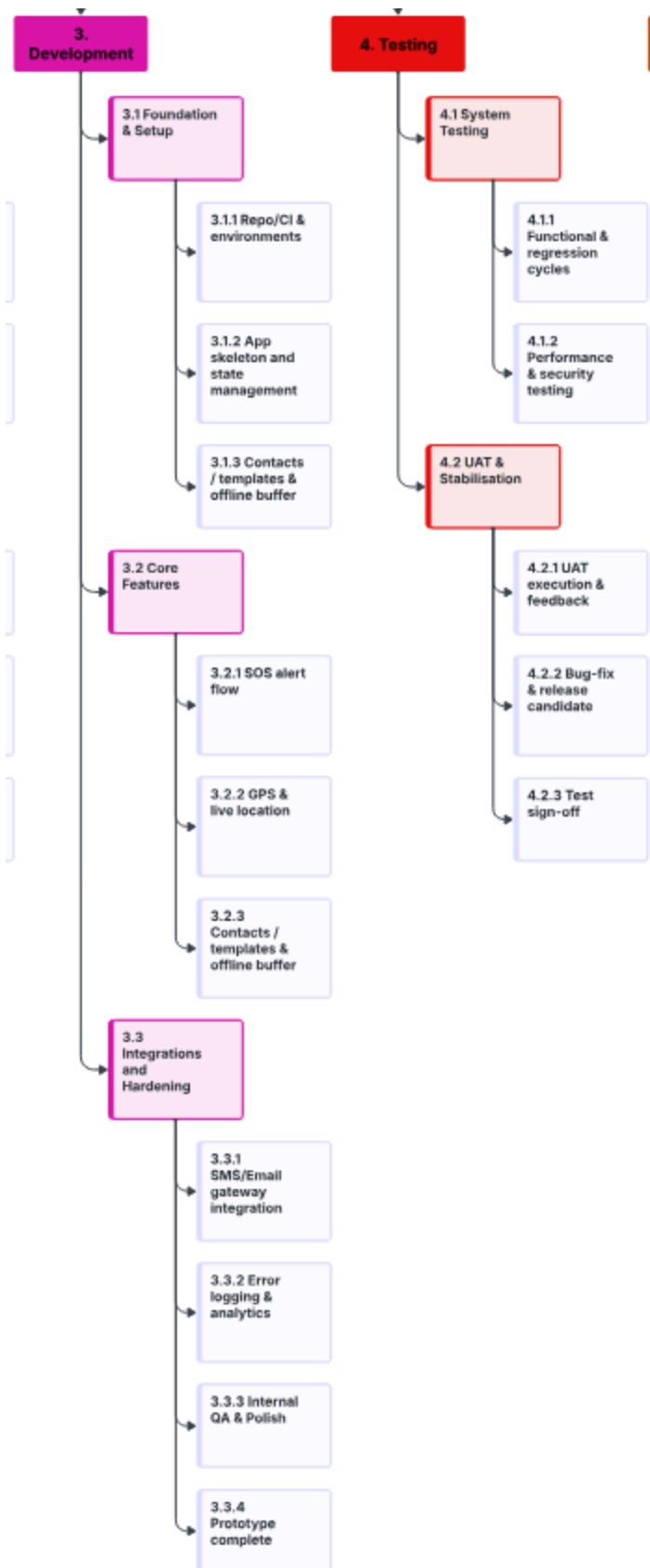
The Project Manager is officially named and given the express power to oversee the project in this crucial administrative deliverable. It will outline the boundaries of their decision-making authority, including their capacity to manage the project schedule, make daily team management decisions, and authorize expenditures up to a specific amount without executive approval. The project's steering committee or other important stakeholders must formally be notified of and approve any modifications that impact the project's scope, budget, or important dates **(Gillis, 2025)**.

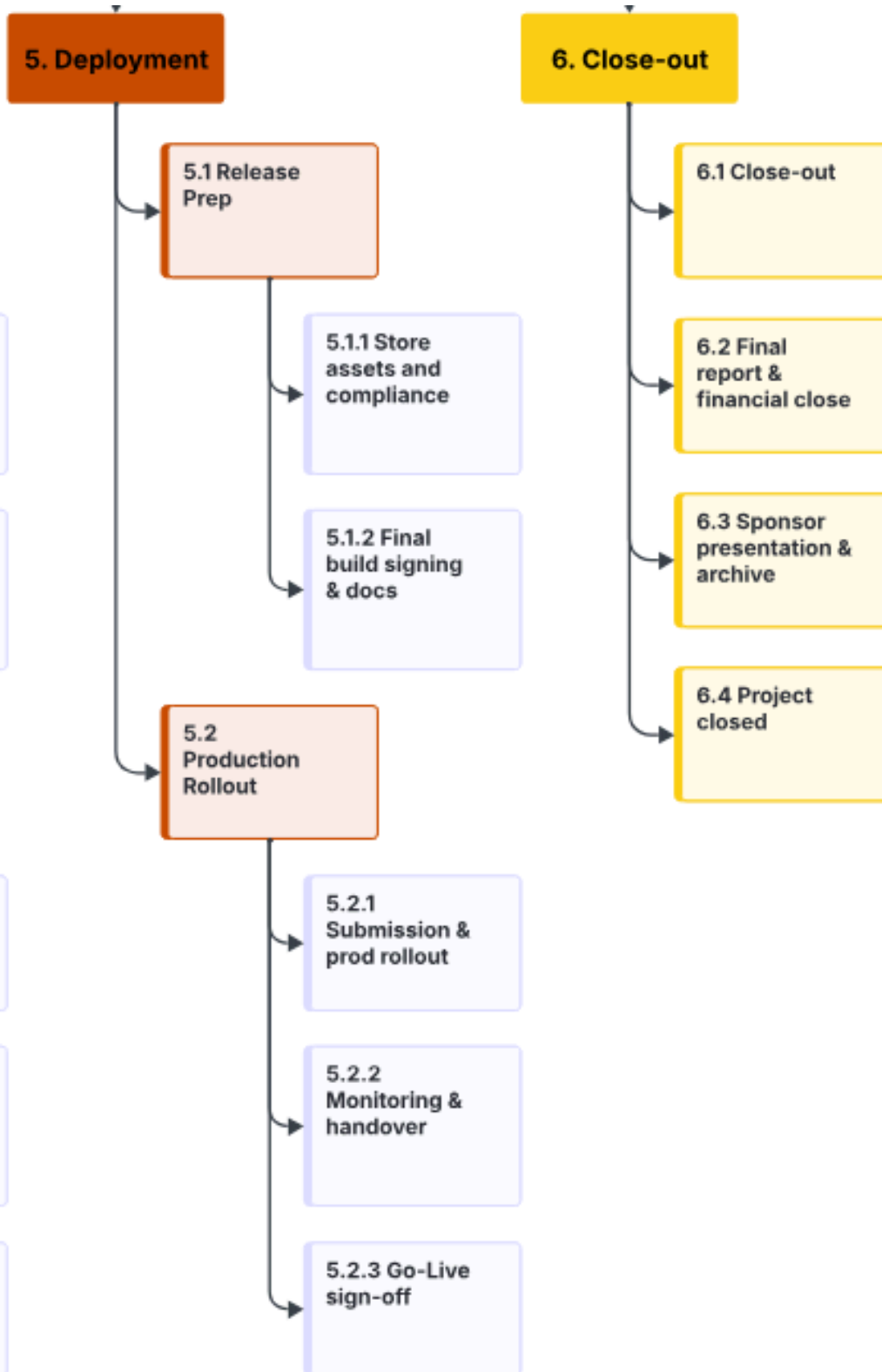
Work Breakdown Structure Diagram



Section Breakdown (For Ease of Marking)







MS Project Solution

See *GitHub* for full project solution:

<https://github.com/ethan-edwards102/INSY6212-Assignment-2>

Resource Assignments

Name	Role
Dylan Olsen	UI/UX Designer
Tyreece Sewbaran	Software Developer
Ndumiso Jwara	DevOps Engineer
Ethan Edwards	Project Manager
David Foster	QA Tester

Summary

WBS	Name	Duration	Start	Finish
0	Emergency SOS App	225 days	Thu 1/1/26	November 12, 2026 8:00 AM
1	Planning	30 days	Thu 1/1/26	February 11, 2026 5:00 PM
1.1	Requirements & Scope	15 days	Thu 1/1/26	January 22, 2026 8:00 AM
1.1.1	Stakeholder & scope workshops	7 days	Tue 10/7/25	January 9, 2026 5:00 PM
1.1.2	Requirements & NFRs baseline	8 days	Mon 1/12/26	January 21, 2026 5:00 PM
1.1.3	Planning Go/No-Go (milestone)	0 days	Wed 1/21/26	January 22, 2026 8:00 AM
1.2	Plans & Baselines	15 days	Thu 1/22/26	February 11, 2026 5:00 PM
1.2.1	Management plans (schedule/cost/risk/comms)	8 days	Thu 1/22/26	February 2, 2026 5:00 PM
1.2.2	WBS, dependencies & set baselines (MSP)	7 days	Tue 2/3/26	February 11, 2026 5:00 PM

2	Design	50 days	Thu 2/12/26	April 23, 2026 8:00 AM
2.1	UX & Prototypes	25 days	Thu 2/12/26	March 18, 2026 5:00 PM
2.1.1	UX research & wireframes	12 days	Thu 2/12/26	February 27, 2026 5:00 PM
2.1.2	Design system & clickable prototype	13 days	Mon 3/2/26	March 18, 2026 5:00 PM
2.2	Technical Design	25 days	Thu 3/19/26	April 23, 2026 8:00 AM
2.2.1	Solution architecture (app/backend/auth)	12 days	Thu 3/19/26	April 3, 2026 5:00 PM
2.2.2	API & data model (+ NFR specs)	13 days	Mon 4/6/26	April 22, 2026 5:00 PM
2.2.3	Design approved (milestone)	0 days	Wed 4/22/26	April 23, 2026 8:00 AM
3	Development	90 days	Thu 4/23/26	August 27, 2026 8:00 AM
3.1	Foundation & Setup	30 days	Thu 4/23/26	June 3, 2026 5:00 PM
3.1.1	Repo/CI & environments	10 days	Thu 4/23/26	May 6, 2026 5:00 PM

3.1.2	App skeleton & state management	10 days	Thu 5/7/26	May 20, 2026 5:00 PM
3.1.3	Security baseline & telemetry	10 days	Thu 5/21/26	June 3, 2026 5:00 PM
3.2	Core Features	30 days	Thu 6/4/26	July 15, 2026 5:00 PM
3.2.1	SOS alert flow	10 days	Thu 6/4/26	June 17, 2026 5:00 PM
3.2.2	GPS & live location	10 days	Thu 6/18/26	July 1, 2026 5:00 PM
3.2.3	Contacts/templates & offline buffer	10 days	Thu 7/2/26	July 15, 2026 5:00 PM
3.3	Integrations & Hardening	30 days	Thu 7/16/26	August 27, 2026 8:00 AM
3.3.1	SMS/Email gateway integration	10 days	Thu 7/16/26	July 29, 2026 5:00 PM
3.3.2	Error logging & analytics	10 days	Thu 7/30/26	August 12, 2026 5:00 PM
3.3.3	Internal QA & polish	10 days	Thu 8/13/26	August 26, 2026 5:00 PM
3.3.4	Prototype complete (milestone)	0 days	Wed 8/26/26	August 27, 2026 8:00 AM

4	Testing	30 days	Thu 8/27/26	October 8, 2026 8:00 AM
4.1	System Testing	15 days	Thu 8/27/26	September 16, 2026 5:00 PM
4.1.1	Functional & regression cycles	8 days	Thu 8/27/26	September 7, 2026 5:00 PM
4.1.2	Performance & security testing	7 days	Tue 9/8/26	September 16, 2026 5:00 PM
4.2	UAT & Stabilization	15 days	Thu 9/17/26	October 8, 2026 8:00 AM
4.2.1	UAT execution & feedback	8 days	Thu 9/17/26	September 28, 2026 5:00 PM
4.2.2	Bug-fix sprint & release candidate	7 days	Tue 9/29/26	October 7, 2026 5:00 PM
4.2.3	Test sign-off (milestone)	0 days	Wed 10/7/26	October 8, 2026 8:00 AM
5	Deployment	15 days	Thu 10/8/26	October 29, 2026 8:00 AM
5.1	Release Prep	5 days	Thu 10/8/26	October 14, 2026 5:00 PM
5.1.1	Store assets & compliance	3 days	Thu 10/8/26	October 12, 2026 5:00 PM

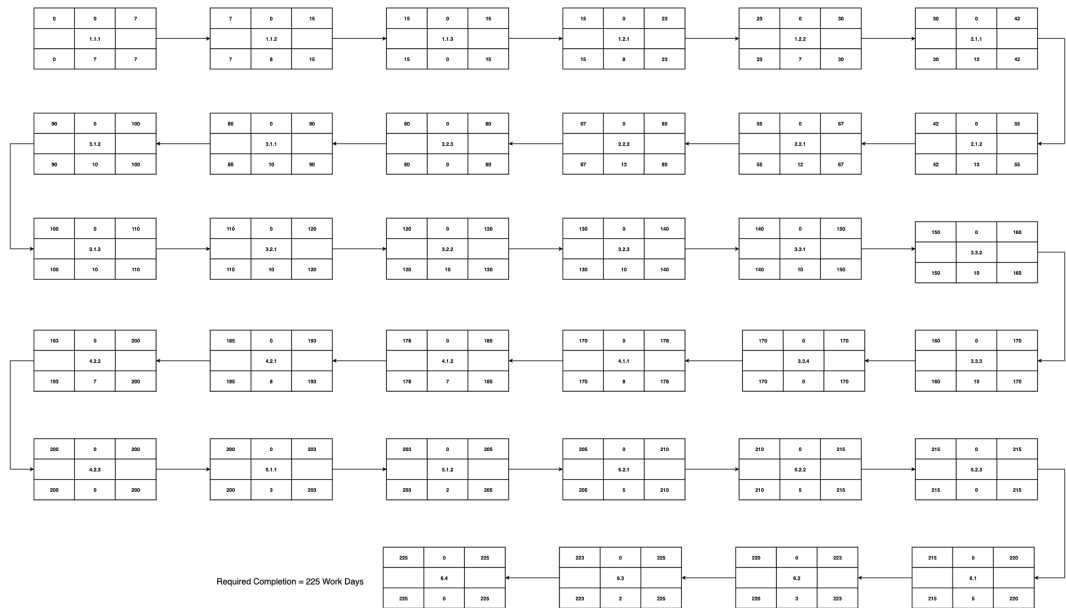
5.1.2	Final build signing & docs	2 days	Tue 10/13/26	October 14, 2026 5:00 PM
5.2	Production Rollout	10 days	Thu 10/15/26	October 29, 2026 8:00 AM
5.2.1	Submission & prod rollout	5 days	Thu 10/15/26	October 21, 2026 5:00 PM
5.2.2	Monitoring & handover	5 days	Thu 10/22/26	October 28, 2026 5:00 PM
5.2.3	Go-Live sign-off (milestone)	0 days	Wed 10/28/26	October 29, 2026 8:00 AM
6	Project Closure	10 days	Thu 10/29/26	November 12, 2026 8:00 AM
6.1	Close-out	5 days	Thu 10/29/26	November 4, 2026 5:00 PM
6.2	Final report & financial close	3 days	Thu 11/5/26	November 9, 2026 5:00 PM
6.3	Sponsor presentation & archive	2 days	Tue 11/10/26	November 11, 2026 5:00 PM
6.4	Project closed (milestone)	0 days	Wed 11/11/26	November 12, 2026 8:00 AM

Network Diagram

KEY

ES	Slack	EF
	X	
LS	Duration	LF

Project Start at 0



Critical Path:

1.1.1 → 1.1.2 → 1.1.3 → 1.2.1 → 1.2.2 → 2.1.1
→ 2.1.2 → 2.2.1 → 2.2.2 → 2.2.3 → 3.1.1 → 3.1.2
→ 3.1.3 → 3.2.1 → 3.2.2 → 3.2.3 → 3.3.1 → 3.3.2
→ 3.3.3 → 3.3.4 → 4.1.1 → 4.1.2 → 4.2.1 → 4.2.2
→ 4.2.3 → 5.1.1 → 5.1.2 → 5.2.1 → 5.2.2 → 5.2.3
→ 6.1 → 6.2 → 6.3 → 6.4

KEY

ES	Slack	EF
	X	
LS	Duration	LF

WBS

Activity

1.1.1 Stakeholder & scope workshops

1.1.2 Requirements & NFRs baseline

1.1.3 Planning Go/No-Go (milestone)

1.2.1 Management plans (sched/cost/risk/comms)

1.2.2 WBS, dependencies & set baselines (MSP)

2.1.1 UX research & wireframes

2.1.2 Design system & clickable prototype

2.2.1 Solution architecture (app/backend/auth)

2.2.2 API & data model (+ NFR specs)

2.2.3 Design approved (milestone)

3.1.1 Repo/CI & environments

3.1.2 App skeleton & state management

3.1.3 Security baseline & telemetry

3.2.1 SOS alert flow

3.2.2 GPS & live location

3.2.3 Contacts/templates & offline buffer

3.3.1 SMS/Email gateway integration

3.3.2 Error logging & analytics

3.3.3 Internal QA & polish

3.3.4 Prototype complete (milestone)

4.1.1 Functional & regression cycles

4.1.2 Performance & security testing

4.2.1 UAT execution & feedback

4.2.2 Bug-fix sprint & release candidate

4.2.3 Test sign-off (milestone)

5.1.1 Store assets & compliance

5.1.2 Final build signing & docs

5.2.1 Submission & prod rollout

5.2.2 Monitoring & handover

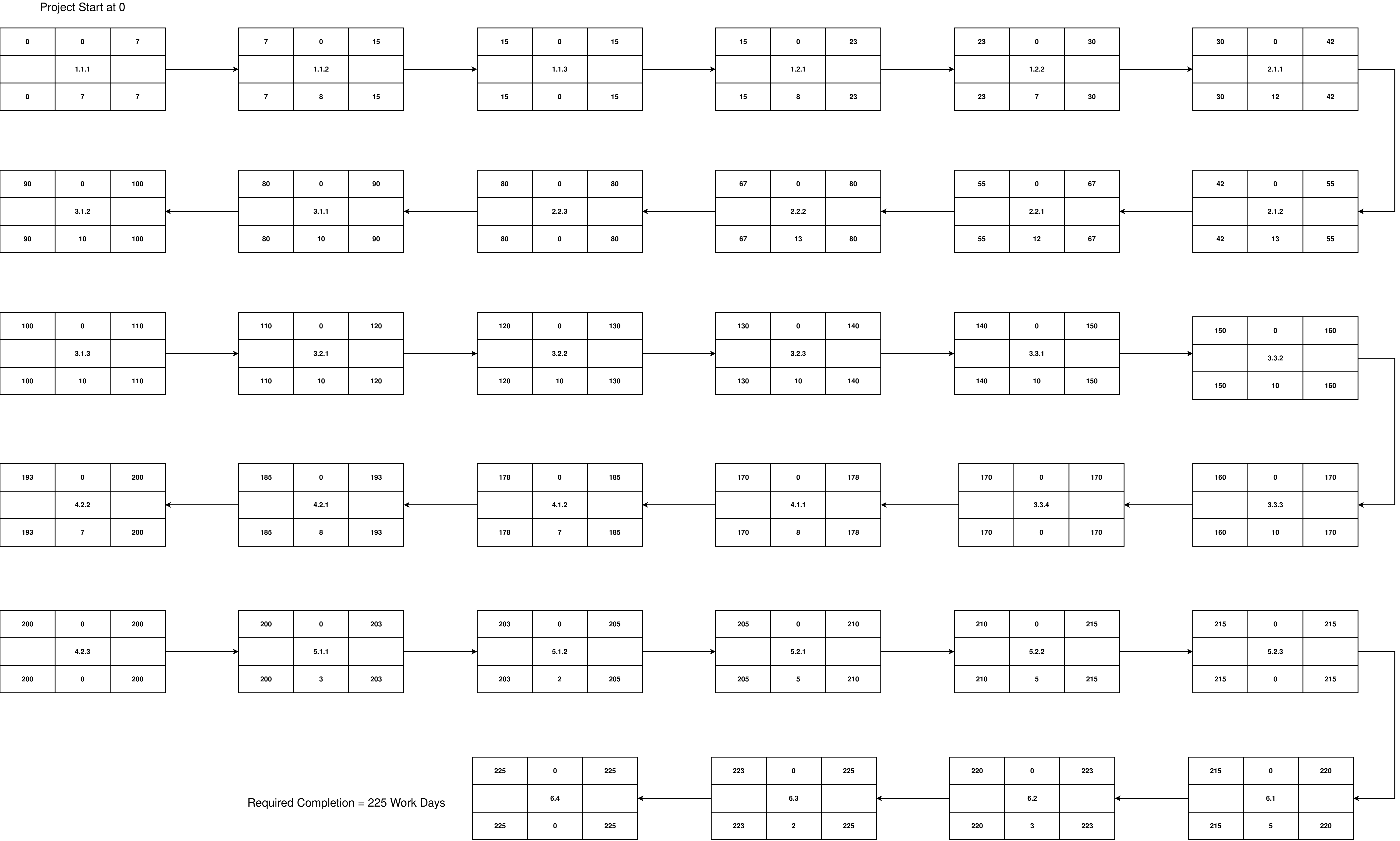
5.2.3 Go-Live sign-off (milestone)

6.1 Close-out

6.2 Final report & financial close

6.3 Sponsor presentation & archive

6.4 Project closed (milestone)



1930200

1937200

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1858193

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1708178

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2100215

2105215

2150215

2150215

2250225

2250225

2230225

2232225

2200223

2203223

2150220

2155220

Required Completion = 225 Work Days

Critical Path:...

GitHub Issues

Labels

Milestones

New issue

☐ Open 7

Closed 0

Author ▾ Labels ▾ Projects ▾ Milestones ▾ Assignees ▾ Newest ▾

☐

Project Closure

#7 · ST10247775 opened now

☐

Deployment

#6 · ST10247775 opened 1 minute ago

☐

Testing

#5 · ST10247775 opened 2 minutes ago

☐

Development

#4 · ST10247775 opened 3 minutes ago

☐

Design

#3 · ST10247775 opened 4 minutes ago

☐

Planning

#2 · ST10247775 opened 4 minutes ago

☐

Emergency SOS App

#1 · ST10247775 opened 1 hour ago

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Warp Development. 2024. *Cross-Platform App Development: Advantages and Disadvantages*. 28 June 2024. [Online] Available at: <https://www.warpdevelopment.com/cross-platform-app-development-advantages-and-disadvantages/> (Accessed 06 October 2025).